Lesson: Python Imports

1. The Core Idea

When you run a Python file, you start with:

- **Built-ins already loaded**
- **Everything else must be imported**

So your script is essentially:

```
[ built-in stuff ] + [ your code ] + [ anything you import ]
```

1. Why Some Things Need Import (and Some Don't)

✅ No import needed

These are **built-in functions and types**:

```python
x = list([1, 2, 3])
print(len(x))
```

Why?
Because they come from Python's **built-in namespace** (automatically loaded).

Examples:

- `list`
- `dict`
- `print`
- `len`
- `int`, `str`, etc.

These are part of Python itself—no import required.

❗ **Import required**

Now look at this:

```python
sleep(2)   # ❌ ERROR
```

This fails because `sleep` is **not built-in**.

You must explicitly import it:

```python
from time import sleep

sleep(2)   # ✅ works
```

Why?
Because `sleep` lives inside the **time module**, not in the built-ins.

2. Where Python Looks for Things

When Python sees a name like sleep, it searches in this order:

1. Your file (variables/functions you defined)
2. Built-ins
3. Imported modules

If it doesn't find it → **NameError**

3. Types of Imports (Single File Usage)

A. Import the whole module

Python Run

```python
import time

time.sleep(2)
```

- Clear and explicit
- Good for beginners

B. Import specific function

Python Run

```python
from time import sleep

sleep(2)
```

- Cleaner syntax
- Slightly less explicit

C. Import with alias

Python Run

```python
import time as t

t.sleep(2)
```

- Useful for long names

4. Why Python Is Designed This Way

Python forces you to import things for 3 main reasons:

1. Performance
It doesn't load everything by default (keeps startup fast)

2. Clarity
You can see dependencies at the top of your file

3. Namespace control
Prevents name conflicts

Example problem if everything were auto-loaded:

```python
sleep()  # Which sleep? From where?
```

5. How to Find Documentation for Imports

This is a critical skill.

Method 1: Official Docs

Search:

```
python time module
```

You'll find:

- Functions (`sleep`, `time`, etc.)
- Usage examples

Method 2: In Python (very important)

```python
import time

help(time)
```

Or:

```python
dir(time)
```

- `dir()` → shows everything inside
- `help()` → explains it

Method 3: Inspect specific function

```python
from time import sleep

help(sleep)
```

6. Built-ins vs Imported (Clear Comparison)

Feature	Built-in (`list`)	Imported (`sleep`)
Available immediately	✅	❌
Needs import	❌	✅
Comes from	Built-in namespace	Module (like `time`)
Example usage	`list()`	`time.sleep()`

7. Best Practice (Single File)

At the top of your file:

```python
import time
```

or

```python
from time import sleep
```

Then your code:

```python
sleep(1)
print("Done")
```

Keep **all imports at the top**—this is standard practice.

8. Mental Model (Important)

Think of Python like this:

- Built-ins = already in your toolbox
- Modules = tools in storage
- import = grabbing tools you need